Lecture 7: **NP**-completeness II

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NP-completeness of HAMILTONIAN CYCLE

HAMILTONIAN CYCLE
Instance: A graph $G = (V, E)$
Question: Is there a Hamiltonian cycle in G, that is, a cycle passing through each vertex exactly once?

- Step 1. The problem is in **NP**: a Hamiltonian cycle is the certificate.
- Step 2. To show completeness, reduce 3-SATISFIABILITY to HAMILTONIAN CYCLE. We will use the method of “component design”.

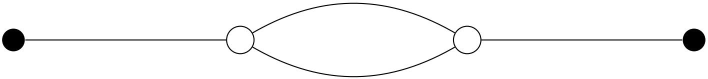
Encoding

Given a CNF-formula f with clauses $C_1, \ldots, C_m$ and variables $x_1, \ldots, x_n$. Encode it as a graph such that the graph has a Hamiltonian cycle if and only if the formula is satisfiable.

What is to be encoded?

- Boolean variables
- a choice between two values (for each variable)
- consistency: all occurrences of any variable get the same truth value
- constraints on the possible values imposed by clauses

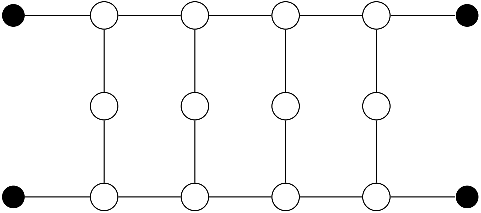
The choice gadget



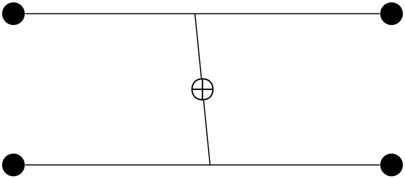
This gadget will allow the Hamiltonian cycle coming through it to pick either left or right “edge”, thus communicating the truth value.

Gadgets are joined to the rest of the graph by the black vertices.

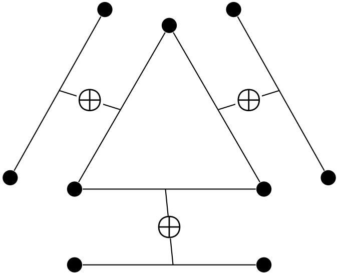
The consistency gadget



It will be connected to the rest of the graph only via the black nodes.
It can be traversed by a Hamiltonian cycle in exactly two ways.



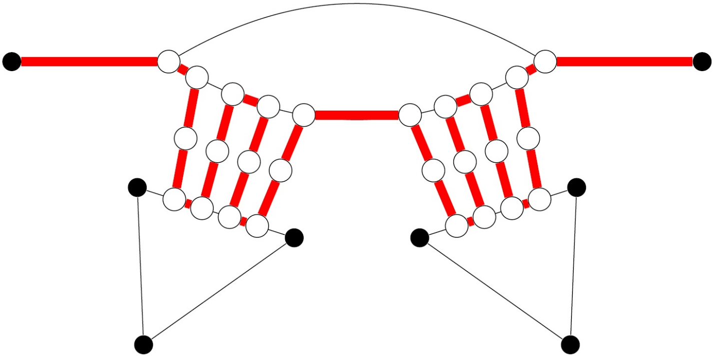
The constraint gadget



- Triangles will correspond to clauses, and their sides to literals. The edges parallel to the sides will be contained in choice gadgets, thus communicating truth values to literals in clauses.
- We will combine the three types of gadgets in such a way that: a side of the triangle is traversed by a Hamiltonian cycle, avoiding its two (black) endpoints, if and only if the corresponding literal is TRUE

$\Rightarrow$ at least one literal in each clause must be TRUE .

Communicating truth value to several triangles

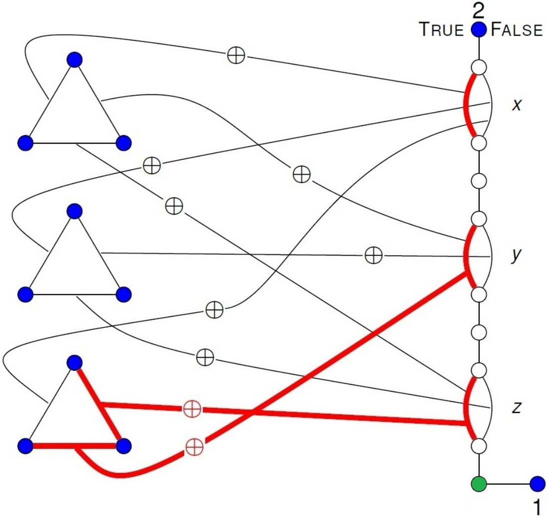


Properties of gadgets

- The choice gadget can be traversed in exactly two ways.
- The internal vertices of the consistency gadget can be traversed in exactly two ways, such that exactly one pair of the external (black) vertices is involved
  - either the black pair on the one side of the gadget, or the black pair on the other side
- Any Hamiltonian cycle (HC) traverses at most two triangle sides using its two (black) endpoints in the constraint gadget:
  - if HC traverses one side using the black endpoints $\Rightarrow$ HC must visit the 2 vertices of this side (one after the other)
  - if HC traverses two sides using the black endpoints $\Rightarrow$ HC must visit all 3 vertices of the triangle (one after the other)
  - if HC traverses three sides using the black endpoints $\Rightarrow$ HC must visit the same vertex of the triangle twice $\longleftrightarrow$ impossible for a HC !

Example construction

$$(x \vee y \vee z) \wedge (\neg x \vee \neg y \vee \neg z) \wedge (\neg x \vee y \vee z)$$



The blue vertices are connected into one big clique.

General construction

- The graph G contains n copies of the choice gadget (one for each variable), connected in a series.
- Add the blue vertices 1,2 and the green vertex.
- G contains m triangles (one for each clause), with a side in the triangle identified with each literal in the clause.
- A side corresponding to literal x_i is connected via the consistency gadget with the "true" side of the corresponding choice gadget a side corresponding to literal $\neg x_i$ is connected to the "false" side.
- If an edge of some choice gadget is connected with several sides of (different) triangles then it is divided into several consecutive edges, each corresponding to a side of a triangle.
  - This is to ensure that the same truth value is communicated to all occurrences of a literal.
- All $3m + 2$ blue nodes are connected into one big clique.

From Hamiltonian Cycle $\longrightarrow$ Truth Assignment

- Every Hamiltonian cycle must use the edge between 1 and the green vertex.
- From 1 via the green vertex, it can only go up through all choice gadgets choosing one side (truth value) in each gadget and traversing the corresponding consistency gadgets on the way.
- It traverses all the triangles in some order (by avoiding the triangle vertices!) and finishes at vertex 2.
- The cycle determines an assignment of truth values to variables.
- Every triangle has at least one side which is traversed avoiding its endpoints (otherwise no HC ! — see above) $\Rightarrow$ the corresponding literal in the clause is TRUE , so all clauses are satisfied.

From Truth Assignment $\longrightarrow$ Hamiltonian Cycle

- Start from 1 to the green vertex and then go through all choice gadgets, each time picking the side corresponding to the truth value of the variable (in the assignment).
- Once this is done, you arrive at vertex 2 at the top of the choice gadget series. Now go through the triangles in arbitrary order, using the sides corresponding to false literals.
- By doing so, you will also visit all unvisited vertices in the choice gadgets. Since all blue vertices of the triangles are fully connected, it is possible to visit every vertex exactly once.
- Once this is done, move and finish at vertex 1.
- This completes the Hamiltonian cycle.

NP-completeness of TSP (Decision Version)

TRAVELING SALESMAN (DECISION VERSION)
Instance: A finite set of cities $\{c_1, c_2, \ldots, c_n\}$, a positive integer distance $d(i, j)$ between each pair (c_i, c_j), and an integer B.
Question: Is there a permutation π of $\{c_1, c_2, \ldots, c_n\}$ such that:

$$\left(\sum_{i=1}^{n-1} d(\pi(c_i), \pi(c_{i+1}))\right) + d(\pi(c_n), \pi(c_1)) \quad \leq \quad B?$$

- Step 1. TSP is in **NP**: a route (permutation) is the certificate
- Step 2. Reduce HAMILTONIAN CYCLE to TSP.

Reduction

Given a graph G with set V of vertices and set E of edges,

- for each vertex v create a city c_v;
- for each pair of distinct $u, v \in V$, set $d(c_u, c_v) = 1$ if $(u, v) \in E$ and $d(c_u, c_v) = 2$ otherwise;
- set $B = |V|$

Then:

- if G has a Hamiltonian cycle then the cycle is a route of cost exactly B
- if there is a route of cost B then it can't use pairs with distance 2 and so goes through edges of G and hence is a Hamiltonian cycle.

NP-completeness of GRAPH 3-COLOURING

GRAPH 3-COLOURING
Instance: A graph $G = (V, E)$
Question: Is there a colouring of the vertices of G in 3 colours such that adjacent vertices are all different colour?

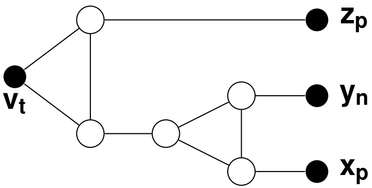
- Step 1. The problem is in **NP**: a 3-colouring is the certificate.
- Step 2. To show completeness, reduce 3-SATISFIABILITY to GRAPH 3-COLOURING.

Encoding

Given a 3CNF-formula f. Encode it as a graph G_f such that the graph has a (proper) 3-colouring if and only if the formula is satisfiable.

- Let's call the three colours ground, true, and false.
- Introduce a 3-clique of designated vertices: v_g, v_t and v_f. By symmetry, assume wlog that they are always coloured ground, true, and false, respectively.
- For each variable x, introduce two vertices, x_p and x_n (for x and $\neg x$), and add all edges between x_p, x_n, and v_g. Hence one of x_p and x_n must be true and the other false.

Encoding cont'd



- For each clause, say $(x \vee \neg y \vee z)$, connect vertices x_p, y_n, and z_p to v_t via the above gadget. That's it.
- Check that, in the above gadget, any combination of colours for x_p, y_n, and z_p, but all false, can be realised.
- Verify that a 3-colouring of G_f can be translated into a satisfying assignment for f, and the other way around.